



Cancer detection app

An Indian student studying in Australia has developed a smartphone app to detect breast cancer

Piss power!

Researchers claim to have come up with the world's first method to charge mobile phones using human urine

A level playing field

We take a look at AMD and Heterogeneous System Architecture Foundation's future plans which aim at making life easier for application developers

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Over the years, we've seen improvements in processor micro-architecture and shrinking of manufacturing processes. Intel has been quite aggressive on that front with its "tick-tock" cadence where the tick represents shrinking of the manufacturing process and tock represents micro-architectural improvements. Although AMD may not be producing powerful CPUs anymore, it has still impressed with its Accelerated Processing Units (APUs) graphics performance. But as anyone who's been following the tech world, you'll know that the future is becoming increasingly mobile-centric. So is the current level of innovation when it comes to processor designs sufficient to take us forward?

Sure there is ARM for the mobile space and Intel and AMD in the x86-x64 space. But going forward is this all the differentiation that's going to prevail? Are we sure that there are absolutely no disruptions to shake up this hegemony?

Before getting onto discussing the future of processor design, let's first look at the problems faced by the current chip design process.

The problem

We've been seeing the processor with integrated graphics for quite some time now. The graphics processing unit (GPU) is ideal for parallel processing, especially when it comes to tasks involving floating-point arithmetic but is not that great at executing single-threaded tasks which may have multiple branches. On the other hand, modern central processing units (CPU) with features such as out-of-order execution are excellent at executing sin-



PlayStation 4 has 8GB of GDDR5 memory and sports an AMD chip

gle-threaded code with branches, however there are shortcomings when it comes to parallel tasks. Using the best of both these units is the main premise of a processor with integrated graphics.

But these two units – CPU and GPU – still have their own separate memory space. A GPU is like a mini system with its own RAM. If an application wants to make use of the GPU segment of the processor, then the CPU needs to carry the data from the system memory to the GPU's memory. This data is then processed in parallel by the GPU and the result has to then be copied back into the system memory. The back and forth transfer increases latency and adds unnecessary wait cycles.

As a result of separate memory pools for the CPU and GPU, the applications need to be coded separately for both of them. The underlying code of a program

cannot be interchanged. This increases complexity for application developers.

Basic working philosophy

The Heterogeneous System Architecture (HSA) Foundation aims to solve the issue faced with current implementation of processors with integrated graphics, of separate memory pools for the CPU and GPU. HSA is an architecture which aims to create an improved processor design, which is able to utilise the serial processing

pro prowess of a CPU and the parallel processing prowess of a GPU and get them working together. This means

that developers don't have to write separate code for the CPU and GPU part of an application execution. Using HSA, applications can create data structures in a single unified address space. This will enable data to be shared between the CPU and the GPU. HSA Foundation aims to come up with open standards to make it easy for developers to program for heterogeneous parallel computing tasks.

The HSA Foundation has major technology companies – including AMD, Qualcomm, ARM, Texas Instruments, Imagination Technologies, MediaTek and Samsung – as its founding members. HSA is not only relevant to processors with integrated graphics, but also to any other co-processing unit which accesses system memory. These co-processing units can be the GPU, hardware engines for specialised tasks such as cryptography, image processing, video transcoding